New belief: Optimal is always better

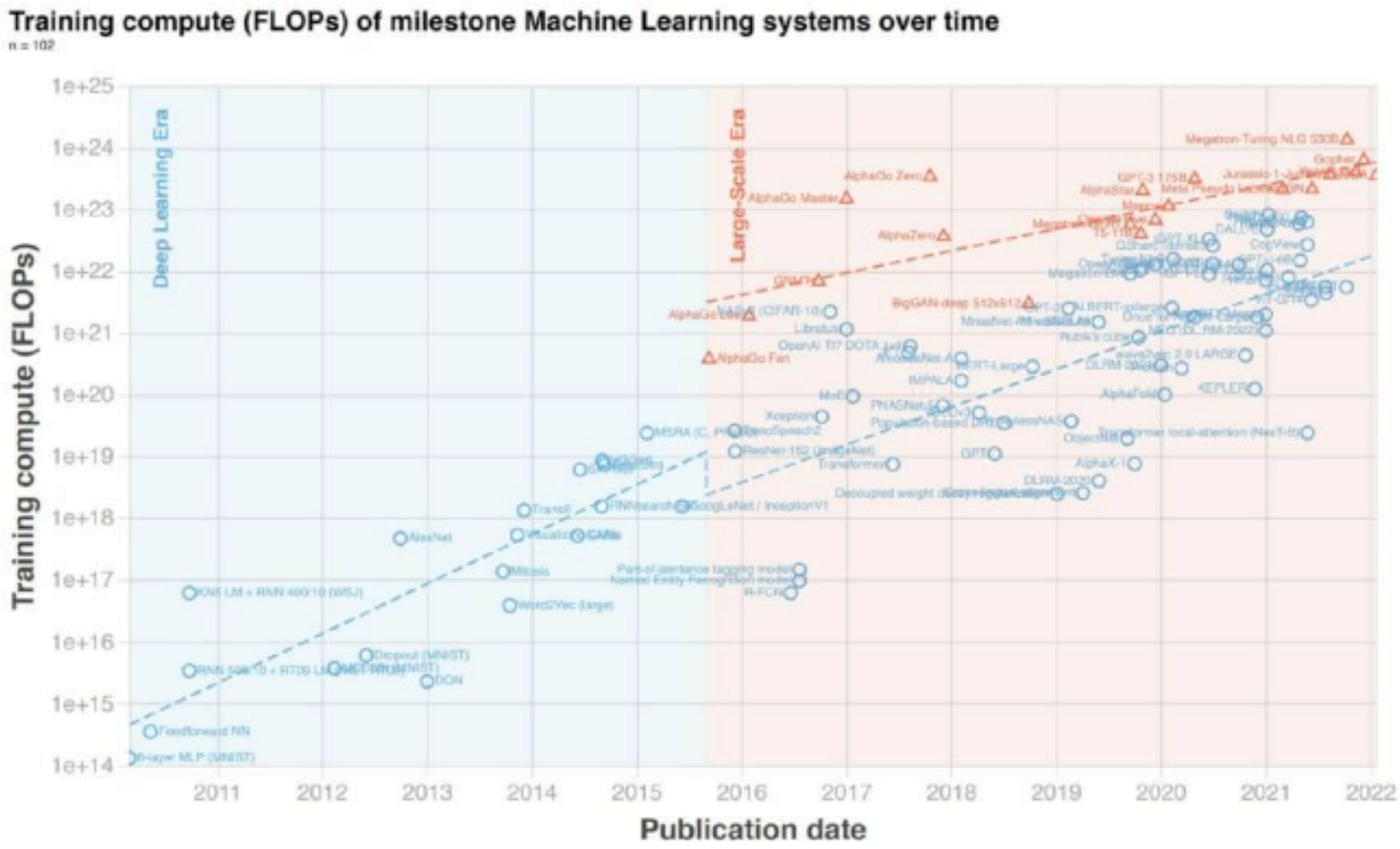

Figure 3: Trends in training compute of $n102$ milestone ML systems between 2010 and 2022. Notice the emergence of a possible new trend of large-scale models around 2016. The trend in the remaining models stays the same before and after 2016.

"Training Compute-Optimal Large Language Models" Hoffman et. Al.

Chinchilla Paper: Some Remarks

Introduction to Chinchilla:

- Chinchilla: A groundbreaking development by DeepMind, reshaping the landscape of large language model training.
- Key Innovation: Prioritizing data scaling rather than increasing the size of the neural network.

- Traditional Belief: Larger models yield better performance – a trend popularized by models like GPT-3
- Chinchilla's Paradigm: Demonstrates that judiciously scaled training data can be more effective than simply expanding model
- Achieves a balance between computational resources and model performance, challenging existing scaling laws.

Resource Efficiency